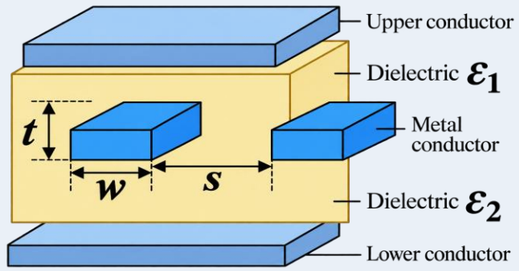
# 1. Physical Context and Sparse Observations

## Geometry and material context




$X, c, S, y_s, b$

**Structured tokenizer**

## Sparse capacitance table

| Spacing $s$ (nm) \ Width $w$ (nm) | 20 | 40 | 80 | 120 | 200 |
|---|---|---|---|---|---|
| 200 | 1.2 | 1.3 | ? | ? | ? ($q_{out}$) |
| 120 | 1.1 | ? | 1.6 | ? | ? |
| 80 | ? | 1.8 | ? ($q_{in}$) | ? | ? |
| 40 | ? | ? | 2.4 | ? | ? |
| 20 | ? | ? | ? | 3.1 | ? |

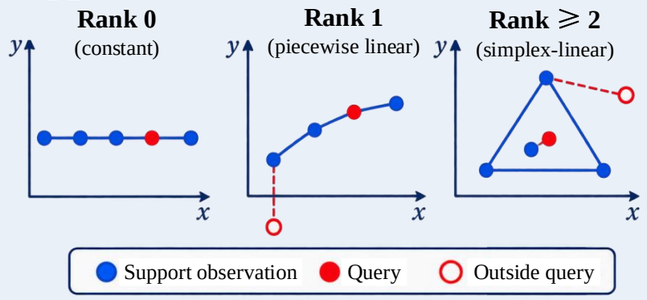
Observed $S$ (value known)
Query $q_{in}$ (inside support)
Query $q_{out}$ (outside support)
Observed support boundary
? Missing value (to predict)

**geometries & materials All known**

$X, S, y_s$

**Support construction**

## Support construction




# 2. Structured Tokens and Shared Representation

## Structured tokens

| Geometry / region | Context $c$ | Observed $C$ ($y/\sigma$) | Seen $m$ | Anchor $b$ | Type |
|---|---|---|---|---|---|
| ($w_1$, $s_1$) | ⋯ | 1.2 / 0.1 | 1 | $b_1$ | Observed |
| ($w_2$, $s_2$) | ⋯ | 1.8 / 0.1 | 1 | $b_2$ | Observed |
| ($w_i$, $s_i$) | ⋯ | 0$_K$ (removed) | 0 | $b_i$ | Query |
| ⋯ | ⋯ | ⋯ | ⋯ | ⋯ | ⋯ |
| Region $\varepsilon_1$ | material | — | — | — | Material |
| Region $\varepsilon_2$ | material | — | — | — | Material |
| ⋯ | ⋯ | ⋯ | ⋯ | ⋯ | ⋯ |

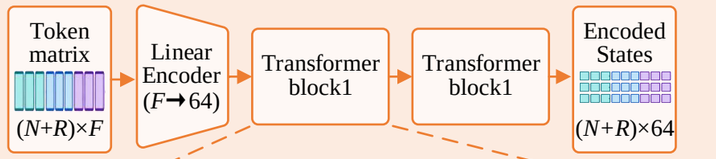
**Query value removed BEFORE embedding (geometry retained)**

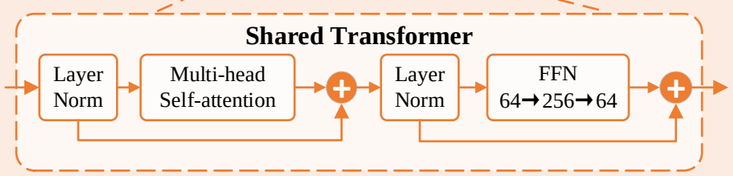
- **Observed token** measurement/value available; Seen m = 1
- **Query token** target value masked before embedding; Seen m = 0
- **Material token** material / dielectric-region context; no capacitance target
- **Masked / N.A.** removed, unavailable, separator, or non-applicable entry

Observed tokens ($|S|\times64$) | Query tokens ($|Q|\times64$) | Materials tokens ($R\times64$)

Token matrix $T$

## Shared embedding and Transformer



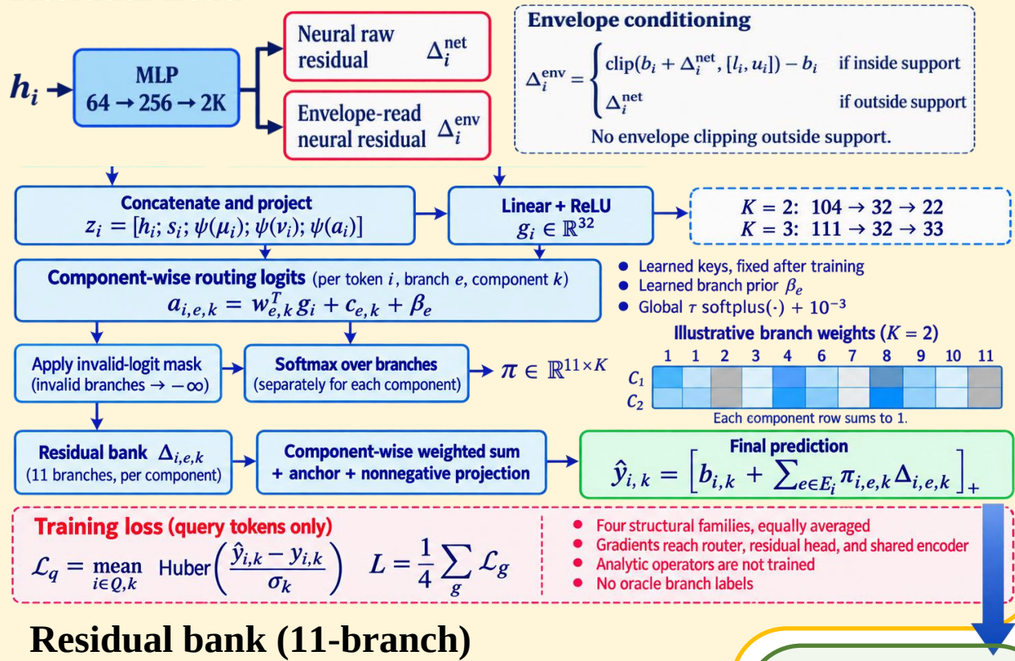

Select query Geometry rows
$H_Q = H^{(2)}[Q,:]$

Query token states
$H_Q = R^{|Q|\times64}$
$h_{i1}$ $h_{i2}$ $h_{i3}$ ... $h_{i|Q|}$

$S$: index set of observed geometries ($|S|$)
$Q$: index set of query geometries ($|Q|$)
$R$: number of material tokens
$N = |S|+|Q|$: total number of geometry tokens
$K$: number of capacitance components (2,3)
$F = 17 + 2K$: token feature dimension

# 3. Learned Residual Bank

## Residual head




## Residual bank (11-branch)

| # | Branch | Recomputed from current support |
|---|---|---|
| 1 | **Anchor** (rule-based) | $\Delta$anchor = 0 |
| 2 | **Neural** (raw) | $r_i = W_r h_i + a_r$, $\Delta$ net = $\sigma \odot r_i$ |
| 3 | **Neural** (envelope-read) | $\Delta$clip = clip($b_i$ + $\Delta$ net, $l_i$, $u_i$) − $b_i$<br>$\Delta$clip = $\Delta$net outside support |
| 4 | **PCHIP** (rule-based) | Single-width, rank-one support; query clamped to observed interval. |
| 5 | **Aligned-width** (rule-based) | Two bracketing widths; other coordinates exactly match the query. |
| 6 | **RBF-GP** (rule-based×6) | $\lambda$= 0.5 , $\rho$=$10^{-5}$ |
| 7 | | $\lambda$= 0.5 , $\rho$=$10^{-4}$ |
| 8 | | $\lambda$= 1 , $\rho$=$10^{-5}$ |
| 9 | | $\lambda$= 1 , $\rho$=$10^{-4}$ |
| 10 | | $\lambda$= 2 , $\rho$=$10^{-5}$ |
| 11 | | $\lambda$= 2 , $\rho$=$10^{-4}$ |
| | | RBF Kernel, Support-mean centering, nonnegative proposals, $\Delta a = \lambda \times$ median positive pairwise support distance |

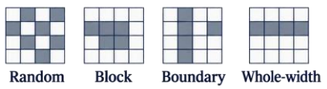
# 4. Predict Missing Entries

| Spacing $s$ (nm) \ Width $w$ (nm) | 20 | 40 | 80 | 120 | 200 |
|---|---|---|---|---|---|
| 200 | 1.2 | 1.3 | 1.5 | 1.7 | 1.9 |
| 120 | 1.1 | 1.4 | 1.6 | 1.8 | 2.0 |
| 80 | 1.6 | 1.8 | 2.1 | 2.3 | 2.5 |
| 40 | 2.0 | 2.2 | 2.4 | 2.7 | 3.0 |
| 20 | 2.3 | 2.6 | 2.9 | 3.1 | 3.4 |

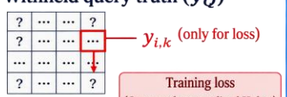
Previously observed (copied exactly)
Predicted (completed)

## Training (query-only supervision)

Masking patterns (examples)




Withheld query truth ($y_Q$)




**Stage 1: Residual warm-up with $[b + \sigma \odot r]_+$**

- Train shared encoder and residual head.
- Use only the neural residual with nonnegative projection.
- No router training.
- Provides a stable initialization.

**Stage 2: Load encoder/head; initialize gate; jointly update all learned parameters.**

- Initial analytic prior gap $\gamma$: 8 for public rules, 2 otherwise.
- Initial global temperature 0.25.
- Jointly train encoder, residual head, and router.
- Warm start at anchor-only prediction.

**Deployment**

- One table forward pass per member.
- Average 3 independently trained members in physical units.
- Restore observed entries exactly.
- Source-only transfer; source scales retained.